Steelmanning LLM Review: Severity Calibration Through Adversarial Self-Challenge

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Large language models can identify methodological concerns in scientific manuscripts but cannot reliably distinguish fatal errors from routine limitations. We formalize this as a severity calibration problem and present a multi-model architecture (graduated dissent) that addresses it through adversarial steelman exchange: separated model instances argue against their own severity ratings before an arbiter renders judgment.

On the SPOT scientific error detection benchmark ($n = 50$ text-detectable papers, externally validated ground truth), the steelman exchange achieves 36% pass@1 with 11.7% retraction-worthy precision, nearly double the precision of alternative architectures, while producing the fewest top-tier flags per paper. On a paired benchmark of 10 retracted papers with 19 non-retracted controls (author-scored with independent multi-model audit), the steelman exchange produces fewer false retraction-worthy classifications on controls (2/19 under independent out-of-family audit) than multi-model pooling (3/19). On SPOT, multi-model pooling improves detection modestly but does not improve severity calibration over rubric-based single-model review. Both observations are consistent with an information-theoretic bound showing that correlated evaluators cannot accumulate independent evidence regardless of ensemble size.

All code, data, model outputs, and a replication harness are released.

Keywords: LLM evaluation; severity calibration; steelman exchange; adversarial self-challenge; multi-model review; error detection.

1. Introduction

A system that produces 15 findings per paper, of which 7% are genuine, does not perform triage; it generates a reading list with a 93% noise floor. The human reviewer must manually evaluate every finding, because the system provides no signal about which findings matter. This is the current state of LLM-based scientific manuscript review.

The practical bottleneck is no longer detection alone. Current frontier models can identify methodological concerns in scientific manuscripts at rates comparable to individual human reviewers.[12] The SPOT benchmark[7] measures whether a model finds a documented error somewhere in its output; the FLAWS benchmark[8] measures error localization. Detection performance continues to improve, but unranked findings are now the practical bottleneck.

The problem is *severity calibration*: the inability to distinguish fatal errors from routine limitations. A paper with a catastrophic statistical error receives the same diplomatic review format as a paper with minor reporting gaps (several strengths, several concerns, a hedged conclusion).[1] This creates a practical bottleneck: as AI-assisted writing increases submission volumes while making surface quality less diagnostic,[17] editorial workflows need automated triage that can rank problems by severity, not merely list them.

We present three contributions:

(1) A multi-model architecture in which separated model instances exchange steelman arguments challenging each other's severity ratings, converting undifferentiated findings into graded severity labels (Section 2).
(2) Empirical evidence from two independent benchmarks showing that the steelman exchange improves detection and nearly doubles within-tier retraction-worthy precision compared to alternative architectures, while multi-model pooling without adversarial exchange provides no severity calibration improvement over rubric-based single-model review (Sections 3–4).
(3) An information-theoretic motivation explaining why pooling correlated evaluators should not be expected to accumulate independent severity evidence, and why adversarial exchange is a structurally plausible remedy (Section 6).

Table 1 summarizes the empirical results.

Table 1. Results across two benchmarks. B3 and graduated dissent (GD) use the same models and arbiter; they differ only in the steelman exchange. Retracted-paper results use out-of-family audit (Grok 4, Mistral Large 2, disjoint from evaluation pipeline).

	Retracted Papers ($n = 10$)		SPOT ($n = 50$)		
Condition	Detection†	FP Rate	pass@1	RW-prec.	RW-yield
B1: Single model	6/10	0/19	24%	n/a	n/a
B2: Single + rubric	9/10	0/19	28%	7.0%	20%
B3: Multi-model pool	7/10	3/19	30%	6.2%	20%
GD: Steelman	10/10	2/19	36%	11.7%	32%

Note: †Out-of-family audit scoring. In-family audit produced 0/19 FP for GD and 2/19 for B3; independent verification revised GD to 2/19 and B3 to 3/19.

2. Method: Multi-Model Evaluation with Steelman Exchange

The graduated dissent protocol[2] treats evaluation as a multi-stage adversarial process.

2.1. *Architecture*

Two **provers** independently review an anonymized manuscript in separated contexts, rating each finding's severity (retraction-worthy, major-revision, minor). A **steelman exchange** then requires each prover to construct the strongest case against their own severity ratings, arguing that their own findings are less severe than initially assessed. An **arbiter** receives all material and applies a conservative severity rubric.

2.2. *Severity rubric*

Every prompt includes the following definitions:

> **Retraction-worthy:** The error means the paper's central conclusions cannot be supported by the data as presented. Not "could be improved" but "fundamentally broken."
>
> **Major-revision:** A real concern that could change conclusions if addressed differently, but does not definitively invalidate them.
>
> **Minor:** Valid criticism applicable to most published papers. Would not change conclusions.

2.3. *Why the steelman exchange targets severity*

Standard multi-model pooling (B3 in our experiments) combines findings from two models but does not challenge severity ratings. If both models share the tendency to over-rate severity on certain error types (a well-documented consequence of RLHF training[1]), pooling reinforces the overestimate.

The steelman exchange disrupts this by requiring each model to argue that its own severity rating is too high. This activates a different prediction target than the default review framing, partially decorrelating the severity assessment from the shared failure modes that produce correlated over-rating.

2.4. *Models*

Table 2. Models used across both benchmarks. Cross-family diversity provides partial decorrelation of failure modes.[11]

Role	Model	Provider	Family
Prover A	GPT-5.4	OpenAI	GPT
Prover B	DeepSeek V3.2	DeepSeek	DeepSeek
Arbiter	Claude Opus 4.6	Anthropic	Claude

2.5. *Budgeted escalation and termination*

The protocol includes a natural termination condition: when inter-prover agreement exceeds a threshold ($\theta_{\text{accept}} = 0.90$), the steelman exchange is skipped and the arbiter receives only the convergent reviews.

In the SPOT evaluation ($n = 50$), the escalation gate fired correctly: 3 papers hit L0 (agreement ≥ 0.90) and skipped the steelman exchange; the remaining 47 escalated to L2 and ran the full adversarial exchange. This provides a stop condition that RLHF-trained review lacks: the protocol terminates when adversarial challenge produces no surviving severity disputes.

3. Benchmark 1: SPOT (Detection and Severity Ranking)

3.1. *Design*

We evaluated all four conditions on the text-detectable subset of the SPOT benchmark,[7] which contains real published papers with human-validated errors. Papers with figure-dependent errors were excluded (our pipeline is text-only), yielding $n = 50$ papers. All papers were anonymized before evaluation.

3.2. *Detection results*

GD achieves 36% pass@1, the highest among evaluated conditions. Multi-model pooling (B3) provides modest improvement over single-model review (+6pp over B1), and the steelman exchange provides a further +6pp beyond pooling. Because our evaluation uses a text-detectable subset ($n = 50$) with different prompting, while the published o3 result used the full multimodal benchmark ($n = 83$), the comparison with o3 should be interpreted as descriptive context rather than a controlled comparison.

3.3. *Severity ranking: the actionability argument*

Pass@1 measures whether the system found the annotated error somewhere in its output, typically a list of 10–15 findings per paper. For triage, the relevant question is not "did it find something?" but "did it tell you this one is the problem?"

GD finds more errors (19 TPs vs. B3's 16) and grades them better: the most true positives in the retraction-worthy tier (9 vs. B3's 6), the fewest total RW flags (77 vs. 97), and nearly double the within-tier precision (11.7% vs. 6.2%). A reviewer who reads only retraction-worthy findings gets the most genuine errors in the fewest flags.

3.4. *Budget-matched controls*

To control for compute budget, two additional conditions were evaluated at the same five API calls as GD ($n = 49$; one paper excluded due to API content filter).

3.5. *Scoring validation*

True positive verification uses a five-layer validation stack: (1) SPOT's author-acknowledged ground truth; (2) SPOT's own LLM-as-judge matching protocol applied verbatim; (3) two adversarial model audits following an 8-step verification

Table 3. Budget-matched conditions on SPOT ($n = 49$). All three use five API calls with different framing for the additional passes.

Condition	Calls	pass@1	RW-prec.	RW-yield	RW/paper
B3+: Neutral reflect	5	22.4%	7.2% (6/83)	19.4%	1.69
B3++: Defend ratings	5	36.7%	6.2% (7/112)	19.4%	2.29
GD: Steelman	5	34.7%	10.5% (8/76)	29.6%	1.55
B3 (3-call ref.)	3	28.6%	5.2% (5/96)	17.2%	1.96

protocol; (4) manual human spot-check of a random sample (5/19 GD TPs, all confirmed); and (5) blinded human verification by a graduate student who independently re-scored all final true-positive decisions without access to condition labels, achieving 100% agreement.

4. Benchmark 2: Retracted Papers (Severity Calibration)

4.1. *Design*

The SPOT benchmark tests detection and severity ranking but does not test false positives on sound papers. We constructed a benchmark of 10 retracted papers with documented methodological errors, paired with 19 non-retracted controls. All retracted papers post-date model training cutoffs. Ground truth was encoded as keyword-decomposed retraction causes. Full details are in Ref. 3.

Detection and false-positive scoring was performed by the first author, verified by in-family audit (1,395 judgments) and out-of-family audit (1,318 findings scored by Grok 4 and Mistral Large 2, disjoint from the pipeline).

4.2. *Results*

Under OOF audit, GD detects all 10 retraction-causing errors (10/10) versus B3's 7/10, and produces fewer false positives (2/19 vs. 3/19). The original 0/19 FP result under in-family audit does not survive independent verification.

The steelman exchange's contribution on this benchmark is not detection but severity calibration: producing fewer false retraction-worthy classifications on controls than multi-model pooling.

5. Cross-Benchmark Analysis

The two benchmarks test different capabilities and tell a consistent story. On SPOT, the steelman exchange's primary effect is grading quality. On retracted papers under independent audit, the primary effect is discrimination. Both serve the same practical function: converting raw model output into ranked severity labels.

The limited effect of multi-model pooling (B3) is equally informative. On SPOT, B3 provides modest detection improvement but no severity calibration improvement: its RW-precision (6.2%) and RW-yield (20%) match rubric-based single-model B2.

6. Why Pooling Does Not Improve Severity Calibration

The empirical pattern admits a formal explanation. This section provides information-theoretic bounds that motivate the observations.

6.1. *Setup*

Let $G \sim p(g \mid x)$ be a generator, $S \sim p(s \mid x, g)$ a selector, and $T := \mathbb{I}\{G = H^\star(X)\}$ the correctness indicator.

Modeling assumption. We assume $S \perp T \mid (G, Z)$ where $Z \in \mathcal{Z}$ indexes shared failure modes.

Theorem 1 (Information Bound via Shared Blind Spots). *Assume* $S \perp T \mid (G, Z)$. *Then:* $I(T; S \mid G) \leq I(T; Z \mid G)$. *If* $T \perp Z \mid G$, *then* $I(T; S \mid G) = 0$.

Lemma 1 (Repeated Self-Critique Bound). *For* k *selectors with* $(S_1, \ldots, S_k) \perp T \mid (G, Z)$ *jointly:* $I(T; A_{1:k} \mid G) \leq I(T; Z \mid G)$.

Under correlated error, k evaluations accumulate subjective confidence without accumulating objective evidence. This is consistent with B3's failure to improve severity calibration on either benchmark.

Proposition 1 (Sufficient Conditions for Positive Evidence). *Assume a selector satisfies* $\Pr(A = 1 \mid T = 0) \leq 1 - \epsilon$ *and* $\Pr(A = 1 \mid T = 1) \geq \delta$ *for some* $\delta > 1 - \epsilon > 0$. *Then acceptance provides* $\log_2 \frac{\delta}{1-\epsilon} > 0$ *bits of evidence about correctness.*

The steelman exchange is designed to partially satisfy this condition, reducing false acceptance of overblown severity classifications.

7. Related Work

Liang et al.[12] found GPT-4 feedback overlapped with human reviewers at 30–39%. SPOT[7] tests detection of verified errors (o3 achieves 18.4% pass@1). FLAWS[8] tests error localization. Pub-Guard-LLM[9] achieves ∼91% on binary retraction classification. None test severity calibration.

Du et al.[10] showed multi-agent debate improves factuality. Huang et al.[4] showed debate performs no better than self-consistency at matched compute. Chen et al.[11] found model family diversity contributes 6.8% of gains. Our architecture differs: models argue against their own positions (steelman), target severity rather than correctness, and preserve disagreement.

Kim et al.[6] document error correlation across 350+ models. Li et al.[13] identified 12 latent biases. Tsui[5] found a 64.5% self-correction failure rate. These motivate the adversarial self-challenge approach.

8. Limitations

(1) **Sample size.** The retracted-paper benchmark comprises $n = 10$ retracted papers with 19 controls. These results motivate larger-scale replication.
(2) **SPOT subset.** Results are on the text-detectable subset ($n = 50$), excluding ∼30 papers with figure-dependent errors.
(3) **Disputed scoring.** One SPOT discordant pair (paper_014) was classified differently by two independent auditors. Strict scoring is reported throughout.
(4) **Retracted-paper scoring.** Initial scoring was performed by the first author, with independent multi-model audit and out-of-family audit verification. The in-family 0/19 FP result does not survive OOF verification (2/19). All audit data are released for independent adjudication.
(5) **Theoretical bounds require falsifiable Z.** The information bound is non-vacuous only when Z is independently constrained.
(6) **Context separation is heuristic.** The steelman exchange partially disrupts error correlation but does not break parameter-level correlations from shared training.
(7) **Cost.** The full pipeline costs approximately \$1.25 per paper.

9. Discussion

All four architectures exceed SPOT's published o3 baseline on text-detectable errors. Detection performance has improved enough that unranked findings are now the practical bottleneck.

The steelman exchange works not by finding more errors but by generating better judgments about errors already found. This is consistent with the theoretical framework: pooling correlated evaluators reinforces shared biases (Lemma 1), while adversarial challenge partially disrupts the correlation (Proposition 1).

To support practical adoption, the complete pipeline (including prompts, scoring code, model configurations, and a browser-based interface) is released as open-source infrastructure.

10. Conclusion

Actionable severity ranking, not raw detection, is the bottleneck for practical LLM-based scientific evaluation. Among the evaluated conditions, the steelman exchange is the only intervention that improves severity calibration across both benchmarks.

On SPOT ($n = 50$), it achieves 36% pass@1 with 11.7% retraction-worthy precision and the fewest top-tier flags per paper. On retracted papers ($n = 10$), GD detects all 10 retraction-causing errors with 2/19 false positives on controls under independent out-of-family audit, compared to B3's 7/10 detection with 3/19 false positives.

All code, data, model outputs, and a replication harness are available at `https://github.com/AndBrilliant/GraduatedDissentBench`.

Data Availability

All code, model outputs, scoring data, and audit reports are publicly available at `https://github.com/AndBrilliant/GraduatedDissentBench`.

Use of AI Tools

This research evaluates large language models as the subject of study. All models used, their roles, and their outputs are documented in the methodology and released in the repository.

Conflicts of Interest

The author is the developer of the graduated dissent protocol and the associated open-source software. The author has no financial interest in the models or API services used. No external funding was received.

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